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RESEARCH ARTICLE

Edge-YOLO: Lightweight Multi-Scale Feature Extraction for Industrial Surface Inspection

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ABSTRACT The increasing complexity of industrial quality control necessitates advanced defect detection systems capable of identifying small-scale surface anomalies with high precision. While deep learning methods, particularly You Only Look Once (YOLO) architectures, have shown promise in object detection, they frequently encounter limitations when detecting minute defects, primarily due to inadequate feature retention in shallow network layers and computational constraints in industrial settings. This study introduces Edge-YOLO, an innovative network architecture that enhances defect detection through a novel edge-enhanced backbone (Edge-backbone). The Edge-backbone systematically strengthens edge feature extraction and retention through three synergistic components: an Edge-Sensitive (EdgeS) module for optimized feature initialization, a Cross-Stage-Partial Edge Enhancement (C3E2) module that integrates edge information across network stages, and a Multi-Scale Dilated Convolution (MSDC) module that efficiently fuses multi-scale features through weight-sharing. These modules work in concert to create a comprehensive edge-aware feature extraction pipeline. Comprehensive evaluation across three industrial datasets demonstrates Edge-YOLO's effectiveness, achieving mean Average Precision (mAP@50) improvements of 2.1%, 3.5%, and 10.3% on NEU-DET, GC10-DET, and GEAR-DET datasets respectively, compared to YOLO11. Notably, Edge-YOLO reduces computational complexity by 69.8% while maintaining competitive accuracy, making it particularly suitable for real-time industrial applications requiring both precision and efficiency.

INDEX TERMS Defect detection, deep learning, edge enhancement, feature extraction, industrial quality control.

I. INTRODUCTION

Defect detection is fundamental to industrial quality control and manufacturing competitiveness. As manufacturing processes become more sophisticated, the requirements for surface defect detection have intensified, as defects can compromise product safety, quality, and longevity, resulting in significant economic losses. The challenges in defect detection are multifaceted. Defects vary widely in form, from microscopic cracks to surface deformations, with many being difficult to detect due to their seamless integration with defect-free surfaces. Environmental factors, such as non-uniform illumination and complex textures, further complicate detection processes, particularly for small-scale defects that require high-resolution imaging and advanced detection algorithms.

Traditional inspection methods face significant limitations. Manual inspection is subjective and prone to fatigue, while conventional machine vision systems lack adaptability to varying conditions due to their reliance on predetermined parameters [1]. These limitations have hindered widespread industrial implementation.

Deep learning technologies have revolutionized defect detection, offering superior real-time performance and reliability. Notable algorithms like YOLO and Faster R-CNN have improved quality control across various industries [2].

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However, challenges persist in detecting small-scale defects, including accuracy limitations, data acquisition difficulties, and computational resource optimization.

To address these challenges, researchers have proposed various strategies to enhance feature extraction capabilities and optimize feature fusion mechanisms. In terms of feature extraction enhancement, attention mechanisms, transformers, and residual connections have been introduced to strengthen the backbone network's feature extraction capability [3], [4], [5]. While these techniques help networks better focus on critical regions, certain limitations persist. For instance, backbone networks lack feature maps that simultaneously contain rich positional and semantic information, limiting network performance in defect detection tasks. Additionally, as network depth increases, enhanced feature information tends to gradually diminish during transmission, particularly affecting small object detection. Finally, some methods introduce numerous additional modules to further enhance feature information, potentially increasing model complexity and reducing inference speed.

Regarding feature fusion optimization, the primary approach involves designing new feature propagation and interaction pathways to improve multi-scale defect detection capabilities [6], [7]. While this approach benefits overall multi-scale defect detection performance, its effectiveness in detecting small-scale defects remains limited, especially under complex background interference. This limitation is particularly pronounced when small-scale defects inherently possess sparse feature information while competing with abundant interference features, making detection increasingly challenging.

In light of the aforementioned challenges, the main contributions of this paper are summarized as follows:

1) We propose Edge-YOLO, a novel network architecture based on YOLO11 for industrial defect detection. This work introduces an innovative edge information enhancement strategy to improve multi-scale small defect detection accuracy. By reinforcing edge information representation at different stages of feature extraction, the proposed method achieves more precise detection of subtle defects.
2) To address the challenge of multi-scale defect detection in industrial scenarios, we introduce three innovative modules:
   - The EdgeS module, serving as the network's entry point, enhances initial edge feature extraction capabilities through a multi-branch structure, establishing a robust foundation for subsequent feature extraction.
   - The C3E2 module, which incorporates an improved CSP structure and edge enhancement mechanism, maintains detailed information while reducing computational complexity during feature extraction.
   - The MSDC module, featuring shared convolutional weights design, reduces parameter count while preserving feature diversity during multi-scale feature fusion.
3) To further enhance model performance, we introduce the LADH (Lightweight Adaptive Detection Head) that combines depth-wise separable convolution (DSConv) with a dynamic anchor generation mechanism. This design effectively reduces computational costs and model parameters while maintaining high-precision bounding box regression and classification performance across various scales of target detection.

The remainder of this paper is organized as follows: Section II reviews related work, with particular emphasis on deep learning-based object detection algorithms and state-of-the-art defect detection methods. Section III provides a detailed description of the YOLO11 model architecture and principles, analyzing the limitations of YOLO11's backbone in multi-scale small object detection. Section IV presents a comprehensive introduction to our proposed Edge-YOLO network and its backbone structure. Section V validates the effectiveness of the proposed method through comparative performance evaluations. Finally, conclusions are drawn in the last section.

II. RELATED WORK

The evolution of defect detection technology can be broadly categorized into three main phases: manual visual inspection, traditional machine learning methods, and deep learning-based approaches. Among these methods, deep learning-based object detection algorithms have garnered particular attention in the field of defect detection. Compared to manual inspection and traditional computer vision methods, deep learning object detection algorithms have distinguished themselves through their superior real-time performance, high accuracy, and excellent robustness, leading to their increasingly widespread adoption in industrial defect detection. Generally, object detection algorithms can be classified into two categories: two-stage and single-stage approaches, differentiated by whether they process detection tasks through two independent steps or a continuous process.

Two-stage detection algorithms, such as Faster Region-based Convolutional Neural Network (Faster R-CNN) [8], employ two specific networks for region proposal and classification/regression tasks respectively. These algorithms utilize VGG [9] or ResNet [10] as backbone networks for image feature extraction. The first stage generates candidate regions through a Region Proposal Network (RPN), while the second stage employs ROI pooling to extract features from candidate regions for object classification and bounding box localization. Although these two-stage methods achieve high detection accuracy, their processing speed is relatively slow due to the requirement for independent network computations on each candidate region.

Single-stage algorithms like YOLO, SSD [11], and CenterNet [12] eliminate the region proposal (RoI extraction) step

present in traditional two-stage methods, employing a single neural network to process the entire task, thereby improving efficiency and reducing complexity. The YOLO series represents the most prominent algorithms among single-stage detectors [13], [14], [15], [16], [17], [18], [19], [20], [21]. YOLO utilizes Convolutional Neural Networks (CNN) to extract image features and directly predicts object bounding boxes and categories from these features, achieving efficient real-time object detection. YOLO11 [22], developed by the authors of YOLOv5 [23] and YOLOv8 [22], introduces the C2PSA (Conditional Channel and Positional Spatial Attention) module, significantly enhancing the model's spatial attention processing capability as a major improvement over YOLOv8. Compared to YOLOv8, YOLO11 supports export formats covering almost all mainstream deep learning frameworks and inference engines, ensuring maximum deployment flexibility across various environments. Generally, industrial defect real-time detection requires maintaining high detection speed while providing sufficient accuracy for practical applications, making single-stage detectors typically more practical in this context.

Benefiting from the rapid development of deep learning-based object detection models, researchers have proposed various improved versions for defect detection applications: Cao et al. [24] introduced YOLOv8-GD, which substantially enhanced feature extraction capabilities by incorporating depth-wise convolution in the backbone and integrating group shuffle convolution with bidirectional feature pyramid networks in the neck architecture. Dai et al. [25] conceptualized the DFN-PSAN model, synthesizing multi-layer convolutions to construct a novel classification network PSAN with pixel-level attention mechanisms to amplify crucial information processing. Fu et al. [26]developed GD-YOLO, engineering efficient D-LAN and lightweight G-LAN modules integrated into the backbone and neck respectively, achieving reduced computational complexity while maintaining detection precision. Han et al. [27] innovated EDN-YOLO, substituting YOLOv5s's original backbone with EfficientVit and introducing decoupled detection heads for optimizing classification and regression tasks separately. Huang et al. [28] proposed ASD-YOLO, significantly improving detection capabilities for multi-scale irregular defects through deformable convolutions, global attention, and context enhancement modules. Li et al. [5] engineered EFD-YOLOv4, introducing convolutional encoding-decoding and FAM modules to address feature alignment through pixel transformation learning. Li et al. [29]developed Cotton-YOLO, utilizing channel attention mechanisms to enhance key feature representation. Li et al. [30] conceived YOLO-CCS, embedding positional information in the backbone to enhance feature extraction and suppress background interference. Ling et al. [31]presented SDD-Net, designing hybrid composite attention mechanisms to improve feature representation capabilities. Liu et al. [32] introduced EMB-YOLO, expanding the receptive field through large kernel shuffle blocks while achieving model lightweight design.

In contrast to these approaches, our work focuses on optimizing the balance between model efficiency and detection performance on edge computing devices. Through in-depth analysis of edge information importance, we enhance the capture and processing of image boundary features, significantly improving detection accuracy for small objects and complex backgrounds without introducing excessive parameters or structural complexity. Furthermore, we improve multi-scale feature fusion methods to ensure effective feature representation for objects of varying sizes while maintaining low computational costs. These optimizations enable our model to provide high-performance object detection capabilities in resource-constrained environments, offering more reliable support for practical applications.

III. BACKGROUND OF YOLO11

This section provides a detailed examination of the theoretical foundation of the baseline model YOLO11. Subsequently, as our primary motivation, we progressively analyze the limitations of YOLO11 and its backbone architecture in extracting small-object features for multi-scale defect detection, followed by an exploration of the fundamental concepts underlying our development methodology.

A. BASIC ARCHITECTURE

As a representative of single-stage detectors, the YOLO series maintains an excellent balance between accuracy and speed, gaining widespread industry recognition. YOLO11, the latest baseline model in the YOLO series, incorporates spatial attention mechanisms that enable the model to focus more effectively on key regions within images, improving detection accuracy for small objects or partially occluded targets. YOLO11 comes in five versions: YOLO11n, YOLO11s, YOLO11m, YOLO11l, and YOLO11x. YOLO11 demonstrates superior accuracy and speed compared to other versions, with real-time detection potential. In our experiments, YOLO11n achieved higher mAP than mainstream baseline models, leading to our selection of YOLO11n as the baseline model.

The YOLO11n algorithm consists of four main components: Input, Backbone, Neck, and Head. As illustrated in Fig. 1, the input section uniformly scales input images to 640×640 and applies mosaic data augmentation to the scaled images.

The backbone component is a feature extraction network based on CSPDarkNet principles, primarily comprising:

- C3k2 (Fast CSP Bottleneck with two convolution layers) SPPF (Spatial Pyramid Pooling-Fast)
- C2PSA (Conditional Channel and Positional Spatial Attention)
- CBS modules (standard convolution with silu activation function)

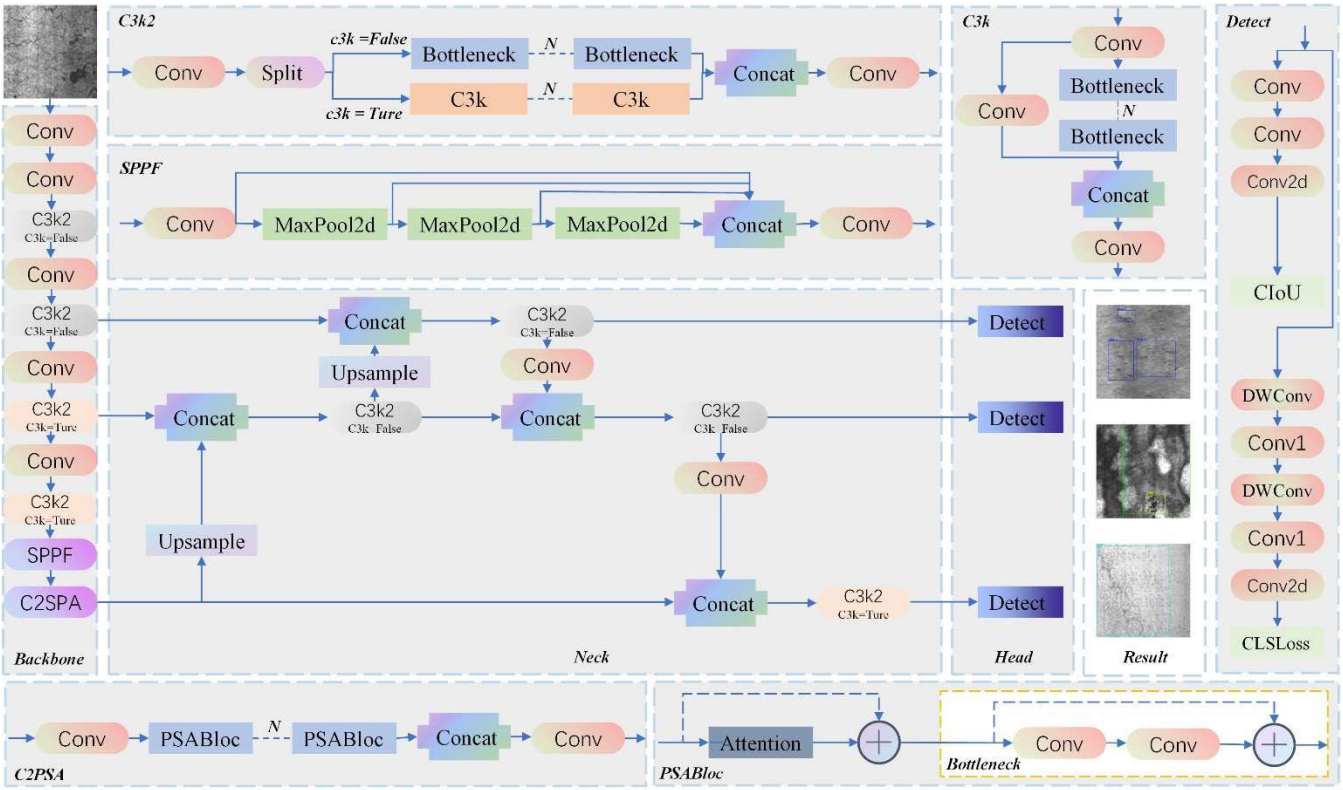


FIGURE 1. The network structure of YOLO11n.

The C3k2-False module inherits the C2F module from YOLOv8, which was inspired by ELAN in YOLOv7, introducing additional gradient branch connections to enrich model gradient flow [18]. The C3k2-True module adds a residual connection to the C2F base, enabling richer feature capture through its more complex structure, resulting in enhanced feature extraction and representation capabilities. The SPPF module transforms feature maps of arbitrary size into fixed dimensions while facilitating local and global feature information fusion. The C2PSA module, a first in the YOLO series, introduces multi-head attention mechanisms, implementing self-attention and feed-forward operations for improved capture of significant image features. The CBS module, a standard convolution block, combines Conv2d, BatchNorm2d, and Silu activation functions. All convolutio layers within these modules perform feature extraction and adjust receptive fields.

B. LIMITATIONS OF CLASSICAL YOLO11 BACKBONE IN MULTI-SCALE SMALL OBJECT DETECTION

The backbone structure encounters several challenges when handling multi-scale small object detection, primarily stemming from key aspects of the network design:

1) Aggressive Downsampling Strategy

The network employs consecutive 2× downsampling from P1 to P5 layers, resulting in rapid reduction of feature map resolution. While this aggressive downsampling approach facilitates the extraction of high-level semantic information, it leads to premature loss of feature information for small objects, particularly in shallower layers, directly constraining the model's ability to effectively capture and identify fine details.

2) Channel Compression Impact

To reduce computational complexity, the network implements channel compression (channel reduction=0.25) in shallow layers, decreasing the number of output channels in each convolutional layer. Although this design significantly reduces computational complexity and memory consumption, it simultaneously weakens the network's capacity to learn detailed features from input images. This limitation is particularly pronounced for small objects, where rich detail features fail to achieve adequate representation, consequently affecting detection accuracy and recall rates.

3) Imbalanced Attention Mechanism Distribution

Notably, the distribution of attention mechanisms across the network is unbalanced. Deep layers are equipped with sophisticated attention modules such as C3k2(True) and C2PSA, while shallow layers remain relatively simple. While this design enhances detection performance for large objects or those with prominent features, it proves inadequate for small objects that typically lack distinct textures or color contrasts. The shallow network layers fail to provide sufficient feature support, impacting overall detection effectiveness. This imbalance becomes particularly evident when handling

multi-scale defects, as accurate detection requires effective feature representation across all levels for different scales of small objects.

To address these limitations, one potential approach involves introducing multi-scale fusion strategies, such as combining deep-layer high-semantic features with shallow-layer high-resolution features through upsampling to enhance small object feature capture. Additionally, adjusting channel compression ratios or incorporating lightweight attention modules in deeper network layers could help preserve more detailed information, thereby improving small object detection performance.

In conclusion, optimizing the backbone structure for small object detection requires consideration not only of computational efficiency but also of ensuring balanced feature extraction across different levels and better preservation and utilization of detailed features for small objects. This optimization demands a comprehensive approach that addresses both architectural design and feature processing strategies.

IV. DEVELOPED METHODS FOR OBJECT DETECTIONOF MULTISCALE DEFECTS

The proposed Edge-YOLO represents a modified version of YOLO11 featuring a specially designed edge-sensitive backbone for multi-scale defect detection. This section presents the intuition and motivation behind designing a lightweight detail-sensitive multi-scale structure for small object detection, followed by a detailed explanation of the Edge-YOLO architecture.

A. INTUITION AND MOTIVATION FOR DESIGNING LIGHTWEIGHT EDGE-SENSITIVE BACKBONE

Enhancing the feature extraction capabilities of deep networks has consistently challenged researchers. In YOLO11, the backbone section serves as the foundation of the entire network, with its performance directly determining the effectiveness of subsequent detection tasks. Improving the backbone is crucial for enhancing YOLO11's detection capabilities, particularly when facing complex and variable industrial defect detection tasks, which serves as the initial motivation for our backbone modifications.

As discussed in Section I, industrial defects typically exhibit characteristics such as mixed scales, extreme aspect ratios, and high confusion potential. Additionally, background interference often exacerbates detection challenges. A critical task in YOLO-driven defect detection is extracting features that match defect characteristics, where features at different levels correspond to different defect sizes. To address these challenges, we propose a novel backbone architecture designed to enhance fine-grained feature capture capability while maintaining computational efficiency.

Compared to normal-sized objects, small objects typically:

- Lack sufficient appearance information
- Possess fewer available features
- Require higher localization precision

It is noteworthy that neural networks characteristically lose object features and position information gradually as network depth increases. The original YOLO11 backbone often struggles to extract sufficiently rich features when processing complex industrial scenes. Therefore, enhancing feature extraction capabilities for small objects in the early backbone stages becomes crucial for improving overall network detection accuracy.

The key to improving the backbone's small object detection capability lies in enhancing sensitivity to feature details. Currently, edge detection and attention mechanisms are widely employed as effective feature enhancement methods. By introducing Sobel operators for explicit edge detection at network entry points while embedding improved attention mechanism modules in subsequent layers, critical features for current tasks can be selectively extracted from vast information. However, previous research has shown that the performance of feature enhancement modules heavily depends on their design structure and integration form with deep networks.

Simultaneously, reducing the computational burden of feature extraction remains a long-standing challenge. While the importance of reducing backbone computational complexity cannot be ignored, attention mechanisms typically increase model structural complexity and lead to complicated computational processes. Implementing lightweight feature enhancement mechanisms in the backbone requires careful consideration.

Therefore, to improve multi-scale defect detection performance, developing a new backbone network that enhances feature extraction capabilities while maintaining low computational complexity becomes essential. This development necessitates a balanced approach between detection accuracy and computational efficiency.

B. ARCHITECTURE OF EDGE-SENSITIVE BACKBONE

The Edge-backbone incorporates Sobel edge detection concepts into the backbone structure, enabling enhanced edge information feature extraction in the lower-level backbone paths. The choice of the Sobel operator over other edge detection methods (such as Canny, Roberts, or Laplacian) is based on both theoretical advantages and empirical results. First, the Sobel operator provides directional gradient information through separate horizontal and vertical components, allowing for more nuanced edge feature extraction. Second, its computational efficiency and simple 3×3 kernel structure make it particularly suitable for deep neural networks, minimizing computational overhead while maintaining effectiveness. Our experimental results validate this choice: the Sobel operator achieves the highest mAP@50 of 80.5% while maintaining low computational complexity (1.9 GFLOPs), outperforming alternatives such as Prewitt (75.8%), Canny (79.4%, with higher complexity at 2.8 GFLOPs), and Laplacian (78.1%). Furthermore, unlike

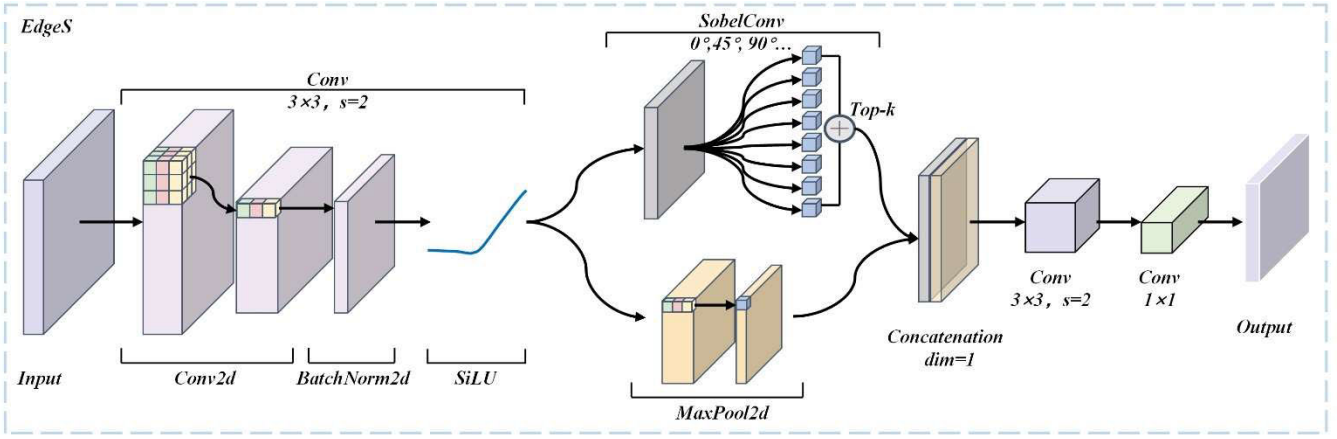


FIGURE 2. The structure of EdgeS.

more complex operators like Canny that involve multiple processing steps, the Sobel operator is fully differentiable and naturally integrates with the network's backpropagation process. This represents one of the primary innovations of our research. This section introduces the developed Edge-backbone architecture and discusses its characteristics.

The proposed Edge Enhanced Backbone (Edge-backbone) structure is illustrated in Fig. 6. Edge-backbone inherits the bottom-up pathway of traditional CSPNet at the structural level while embedding specially designed EdgeS, C3E2, and MSDC modules to replace conventional Conv, Bottleneck, and SPPF modules. Additionally, it maintains YOLO11's concatenation operations, C2PSA modules, and Conv modules in the bottom-up pathway for feature integration and transformation.

The EdgeS module, as shown in Fig. 2, implements a sophisticated multi-branch architecture. The initial convolution layer (conv1) employs a large stride (stride = 2) to halve input feature map dimensions while transforming the channel number from inc to hidc, performing initial low-level spatial feature extraction. The Sobel edge detection branch (sobel_branch) implements the classical Sobel operator for gradient detection, enhancing edge information in feature maps and improving object boundary definition and structural understanding. The pooling branch utilizes ZeroPad2d for dimension preservation and implements MaxPool2d with stride=1 and ceil_mode=True, preserving critical local maximum values while maintaining feature map dimensions.

For feature fusion and channel adjustment, EdgeS concatenates Sobel detection and pooling results along the channel dimension, doubling the channel number to hidc * 2. A second convolution (conv2) with stride=2 is applied, followed by a 1×1 convolution (conv3) for final channel adjustment to ouc channels. This design ensures efficient feature extraction while maintaining computational efficiency. The calculation formula of the above process is as follows:

$$F_{EdgeS}(x) = Conv2d(ReLU(Conv2d(x, W_1)), W_2) \quad (1)$$

As illustrated in Fig. 6, the C3E2 module is a highly optimized and powerful neural network component that inherits the efficient CSP (Cross Stage Partial) structure from C3k2 while introducing EdgeE (Edge Information Enhancement Module) to enhance multi-modal information processing capabilities, as depicted in Fig. 3. This module is designed to improve feature extraction and processing efficiency in computer vision tasks, particularly excelling in tasks requiring precise boundary information capture. The C3E2 module inherits from the C3k2 class, which represents an improved CSP Bottleneck structure designed to accelerate computation through two convolution layers while maintaining robust feature extraction capabilities. The core concept of the CSP structure involves splitting input feature maps into two parts: applying deep convolution operations to one part while directly passing or simply transforming the other, before finally concatenating both parts. This design reduces computational complexity while enhancing model expressiveness.

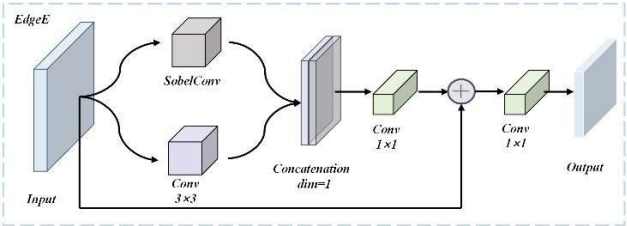


FIGURE 3. The structure of EdgeE.

EdgeE serves as a key component within C3E2, primarily focusing on enhancing edge information in images. It comprises Sobel edge detection and conventional convolution branches, whose outputs are concatenated and fused through a series of 1×1 convolutions for channel adjustment. These feature maps are ultimately added to the original input, creating richer and more accurate feature representations. The calculation formula of the above process is as follows:

$$M_{ch}(F_{in}) = \sigma(W_2(W_1(F_{in})) + W_s(W_t(F_{in}))) \quad (2)$$

$$F_{out}(x) = M_{ch}(F_{in}) \otimes F_{in} \quad (3)$$

wherein:

F_{in} denotes the input feature, W_1, W_2 represent the encoding weights, W_s, W_t represent the transfer weights, σ denotes the activation function, $\otimes$ denotes the matrix multiplication operation.

As depicted in Fig. 4, the MSDC (Multi-Scale Dilated Convolution) module initially employs a 1×1 convolution layer (cv1) to reduce input feature map channels from c1 to c_. This operation reduces subsequent computational complexity and performs dimensionality reduction, enabling the model to focus on essential features. The module introduces a shared convolution layer (share_conv), implementing a 3×3 convolution without bias (bias=False). This convolution layer is reused across different dilation rates, reducing parameter count and improving efficiency. This approach enables effective capture of multi-scale information without significantly increasing computational burden. During forward propagation, the MSDC module first passes input feature map x through cv1 for initial feature representation. For each specified dilation rate, it applies the shared convolution layer while adjusting dilation rates and corresponding padding sizes to ensure consistent output dimensions. Feature maps processed with different dilation rates are collected and concatenated along the channel dimension with the original cv1(x) feature map.Finally, all concatenated feature maps pass through another 1×1 convolution layer (cv2) for channel adjustment, achieving the desired output channel number c2. This process not only completes feature fusion but also provides rich, multi-scale feature representations for subsequent tasks. Through this mechanism, the MSDC module effectively integrates information from different scales, enhancing understanding of both details and global structures. The shared convolution layer enables the module to maintain efficiency while providing powerful feature extraction capabilities. The calculation formula of the above process is as follows:

$$F_{MSDC}(x) = Conv2\, d_{shar}\left(Concat\left[F_{low}, Upsample\left(F_{high}\right)\right]\right) \quad (4)$$

The C2PSA module inherited from YOLO11 is computed as follows:

$$M_{spa}(F_{in}) = \sigma\left(Conv2\, d\left(Maxpool\left(F_{in}\right) + Avgpool\left(F_{in}\right)\right)\right) \quad (5)$$

$$F_{out} = M_{spa}(F_{in}) \otimes F_{in} \quad (6)$$

C. EDGE-BACKBONE WORKFLOW

The Edge-backbone operates through a progressive feature extraction and fusion process, specifically designed for multi-scale object detection tasks. The workflow begins with the EdgeS module for initial feature extraction, which specifically enhances edge information extraction capabilities. Subsequently, multi-scale feature maps are constructed through a series of downsampling operations and C3E2 modules. The shallow network layers (first two C3E2 modules) employ a channel reduction (0.25) strategy to process fine-grained features, primarily preserving position and texture information. The middle and deep network C3E2 modules enable attention mechanisms (True parameter) to enhance semantic feature extraction capabilities.

Following feature extraction, the MSDC module performs multi-scale feature fusion, sharing convolution weights to establish correlations between different scales and effectively combining positional and semantic information. Finally, the C2PSA module serves as a feature enhancement unit, further optimizing feature representation through spatial attention mechanisms to highlight feature responses in critical regions. This progressive workflow forms a comprehensive feature extraction and fusion process, transitioning from fine-grained features to high-level semantic features. The calculation formula of the above process is as follows:

$$F_{final} = C2PSA(MSDC\,(C3E2\,(EdgeS(x)))) \quad (7)$$

Fig. 7 shows the comparison of receptive fields between the traditional backbone (top row) and our proposed Edge-backbone (bottom row) across 10 different network layers. The visualization demonstrates the progressive evolution of receptive fields from layer 1 to layer 10.

The heat maps reveal several key observations:

- Progressive Field Expansion: Both backbones show gradual expansion of receptive fields from shallow to deep layers (left to right), with layer 1 showing focused local attention and layer 10 exhibiting broader field coverage.
- Enhanced Edge Sensitivity: The Edge-backbone demonstrates more structured and defined receptive field patterns, particularly evident in layers 6-10, where the activation patterns show clearer directional sensitivity compared to the traditional backbone.
- Field Distribution: The Edge-backbone exhibits more concentrated and well-defined activation patterns, especially in deeper layers (8)-(10), suggesting more effective feature aggregation and edge information preservation.
- Spatial Coverage: While both architectures show similar progression in receptive field size, the Edge-backbone maintains better feature localization even in deeper layers, as evidenced by the more structured activation patterns in layers 9-10.

This visualization effectively demonstrates how our Edge-backbone architecture achieves enhanced feature extraction capabilities while maintaining precise spatial information throughout the network depth.

D. EDGE-YOLO NETWORK ARCHITECTURE

Fig. 6 illustrates the defect detection architecture of Edge-YOLO. Built upon YOLO11n's basic formulation, the proposed architecture primarily differs in its integration of the Edge-backbone. Edge-YOLO can be interpreted as a model

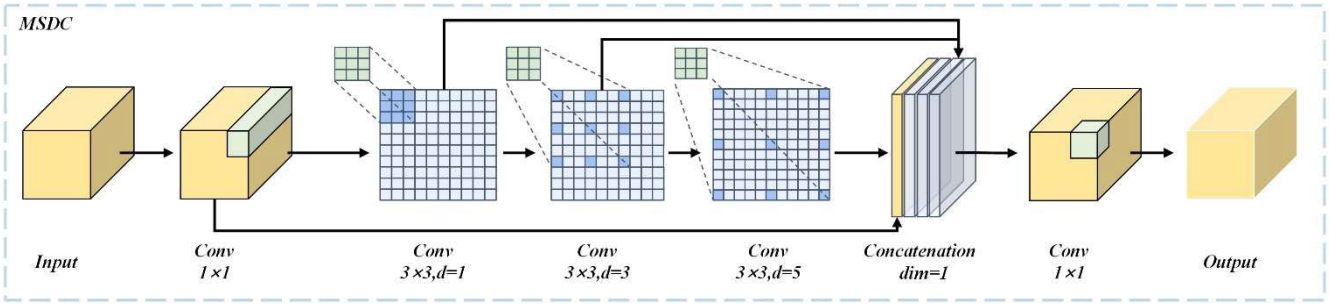


FIGURE 4. The structure of MSDC.

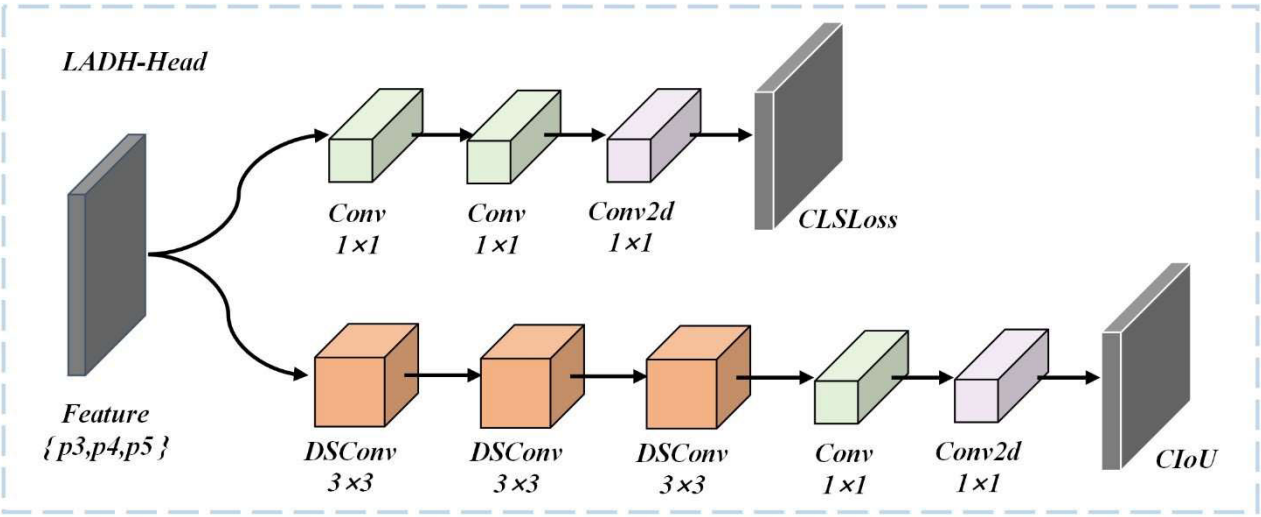


FIGURE 5. The structure of LADH-Head.

composed of stacked components, comprising three main sections: backbone, neck, and head.

The workflow begins with input image processing, where images are uniformly scaled to 640 × 640 and enhanced through mosaic data augmentation. This augmentation process randomly selects two to four original images, applies cropping, rotation, and scaling operations, and merges them into a single enhanced image. This random selection and combination significantly enriches the dataset, particularly beneficial for datasets with limited original samples.

The backbone network progressively extracts feature information through a series of convolution modules. Lower-level convolutions capture basic features rich in spatial position information but lacking semantic content, while deeper convolutions capture high-level features with rich semantic information but reduced spatial precision. The backbone outputs low-, middle-, and high-level feature maps to the neck for feature fusion.

As depicted in Fig. 5, the detection head comprises detection and classification branches for each detection module. The detection branch performs consecutive convolutions on input feature maps to extract bounding box information, while the classification branch processes classification information. Following the approach by Zhang et al. [33], the detection head separates task types across three independent channels. To expand the receptive field and increase IoU branch task parameters, three convolution operations are employed to reduce channel dimension features.

Notably, the architecture replaces traditional 3 × 3 convolutions with 3 × 3 Depthwise Separable Convolutions (DWConv) in each branch to reduce model parameters and improve computational efficiency. DWConv combines depthwise and pointwise convolutions to adjust channel numbers without altering feature map spatial dimensions, effectively mixing information between different channels while maintaining feature extraction capabilities.

V. EXPERIMENTAL EVALUATION

A. DATASET DESCRIPTION

To thoroughly validate the effectiveness and generalization capability of the proposed method, we conducted experiments on two popular public datasets and one self-built dataset: NEU-DET and GC10-DET for steel surface defect detection, and GEAR-DET for gear defect detection. Detailed information is presented in Table 1.

1) NEU-DET: This is a public dataset of steel surface defect detection produced by Northeastern University, which contains six typical surface defects of hot-rolled steel strip: crazing (Cr), inclusion (In), patches

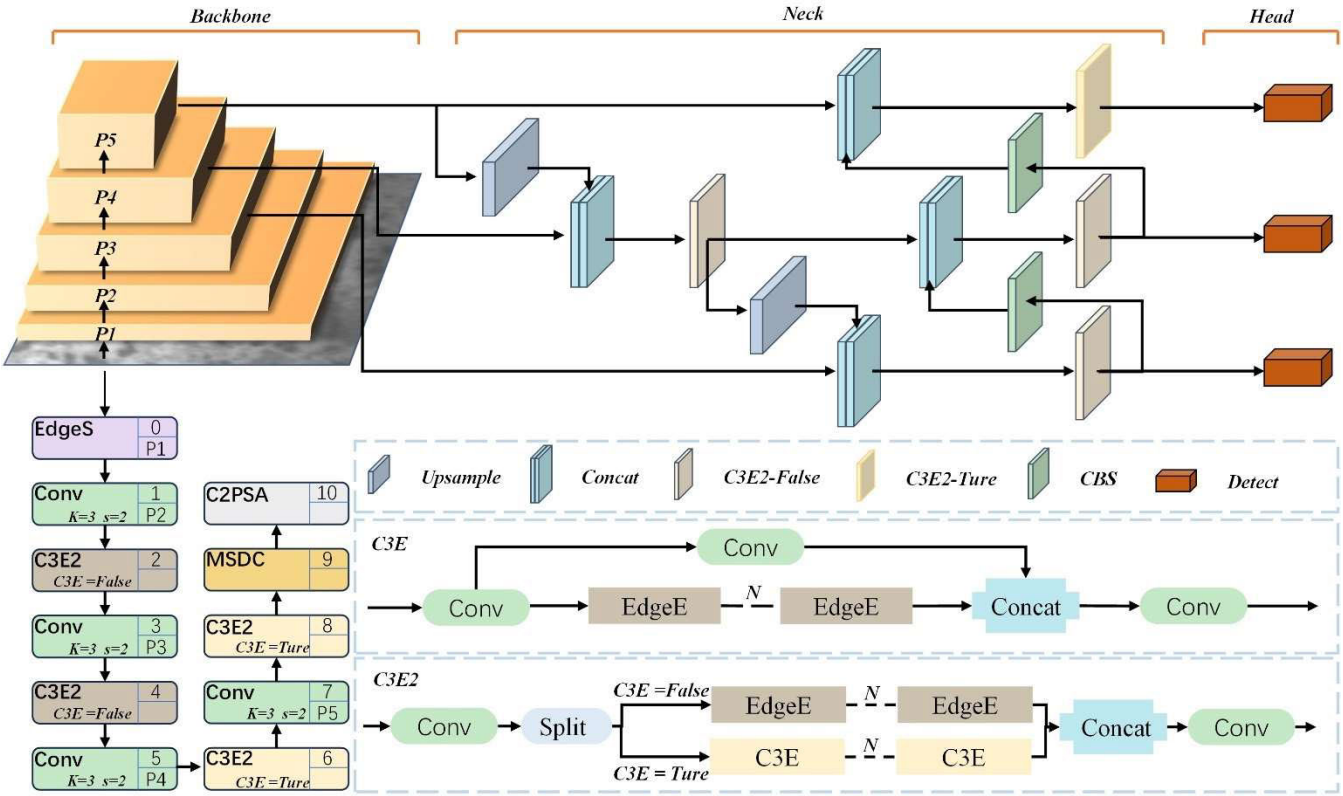


FIGURE 6. The network structure of Edge-YOLO.

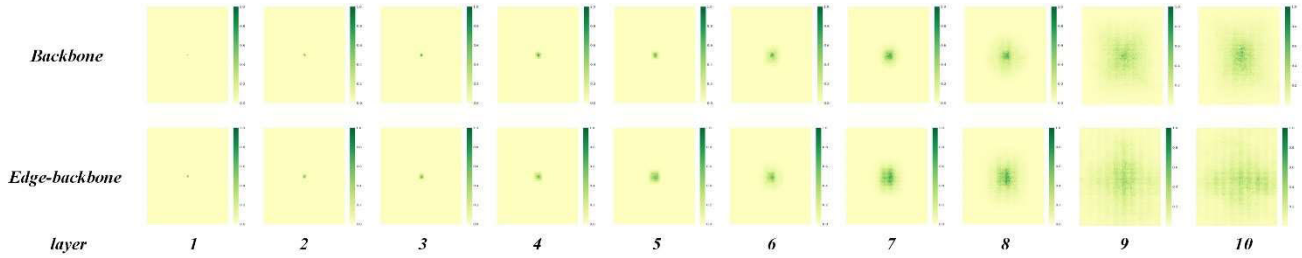


FIGURE 7. The receptive fields of Edge-backbone.

(Pa), pitted surface (Ps), rolled-in scale (Rs) and scratches (Sc). The dataset has a total of 1800 images, 300 images for each defect. For our experimental evaluation, we divided this dataset into 1,260 training images, 360 validation images, and 180 test images.

2) GC10-DET: This is a public dataset of surface defects on steel plates collected in real industrial scenarios. There are ten types of defects: crescent gap (Cg), crease (Cr), inclusion (In), oil spot (Os), punching hole (Pu), rolled pit (Rp), silk spot (Ss), waist folding (Wf), welding line (Wl)and water spot (Ws). After removing unusable data, the dataset comprises 2,294 images. We partitioned the dataset into 1,606 training images, 344 validation images, and 344 test images for our experiments.

3) GEAR-DET: This self-collected gear surface defect dataset contains three typical surface defects: Impact damage (Kp), Tooth surface black skin (Hp1) and Tooth root black skin (Hp2). The dataset consists of 1,397 images, which we divided into 975 training images, 295 validation images, and 127 test images for experimental evaluation.

TABLE 1. Detailed information on the NEU-DET, GC10-DET, and GEAR-DET datasets.

Dataset	Defect type	Total images	Images for train	Images for validation	Images for test
NEU-DET	6	1800	1260	360	180
GC10-DET	10	2294	1606	344	344
GEAR-DET	3	1397	975	295	127

B. EXPERIMENTAL SETUP

The experiments were conducted on a Linux operating system with an AMD EPYC 9754 CPU and NVIDIA GeForce RTX 3090 GPU. We implemented our experiments using the PyTorch deep learning framework (version 2.1.2) with CUDA 11.8 and Python 3.10.

All deep learning methods in our experiments utilized identical hyperparameters. While hyperparameter optimization remains a challenging research endeavor, our focus was on developing a new defect detection model with a specially designed architecture rather than optimizing hyperparameters. The model hyperparameter settings are detailed in Table 2.

TABLE 2. Initialization parameters of our method.

Parameters	Value	Note
Learning rate	0.01	Initial and final learning rate
Momentum	0.937	
Weight decay	0.0005	
Close mosaic	0	Disable mosaic augmentation
Total epochs	700	
Optimizer	SGD	
Batch size	16	

C. EVALUATION METRICS

To comprehensively evaluate model accuracy, we employed commonly used metrics including precision, recall, and Average Precision (AP), with mean Average Precision (mAP) as the primary evaluation metric. These metrics are defined as follows:

$$Precision = TP/((TP + FP)) \tag{8}$$

$$Recall = TP/((TP + FN)) \tag{9}$$

$$AP = \int_0^1 p\,(r)\,dr \tag{10}$$

$$mAP = \frac{1}{n}\sum_{i=1}^{n} AP_i \tag{11}$$

In these evaluation metrics formulas, several key parameters are employed to assess model performance. TP (True Positives) represents correctly identified defects, while FP (False Positives) indicates instances where non-defects are mistakenly classified as defects, and FN (False Negatives) denotes actual defects that the model fails to detect. The precision metric, calculated as TP/(TP+FP), measures the accuracy of positive predictions, while recall, computed as TP/(TP+FN), quantifies the model's ability to identify all actual defects. The Average Precision (AP) is derived by integrating the precision value p(r) over the recall range from 0 to 1, providing a comprehensive measure of detection performance. Finally, the mean Average Precision (mAP) is calculated by averaging the AP values across all n defect categories (AP_i), serving as a holistic metric for evaluating the model's overall detection capability across different defect types.

D. EXPERIMENTAL RESULTS

1) Performance Comparison

To validate the effectiveness of our proposed method, we conducted detection experiments using YOLO11 and Edge-YOLO on the NEU-DET dataset, with results summarized in Table 3. Edge-YOLO demonstrated significant advantages in mAP@50% for Pa, In, and Rs defect categories. Notably, for Pa-type defects, Edge-YOLO achieved a recall value of 94.6%, an 8.6% improvement over the original YOLO11, validating the effectiveness of our edge enhancement strategy in improving model recall.

Among the six defect types, Cr and Ps represent typical small-object defects. The original YOLO11 model achieved mAP@50-95% values of 18.9% and 52.1% for Cr and Ps, respectively. Benefiting from the edge information enhancement of the EdgeS module and the detail preservation capability of the C3E2 module, Edge-YOLO demonstrated superior performance in handling these small-object defects, improving the mAP@50-95% for Ps to 56.4%.

For Sc-type defect detection, Edge-YOLO's mAP@50% was 3.9% lower than YOLO11, primarily due to the MSDC module potentially overemphasizing edge features during multi-scale feature fusion. Since Sc-type defects (scratches) typically present as elongated linear features, excessive edge enhancement may interfere with the model's perception of overall shape.

TABLE 3. Detection performance of YOLO11 and Edge-YOLO on NEU-DET datasets.

	Class	P%	R%	mAP@50/%	mAP@50-95/%
YOLO11	Cr	67.0	30.3	45.5	18.9
	In	78.6	71.4	75.3	40.5
	Pa	82.0	86.0	91.6	56.9
	Ps	92.0	87.1	90.0	52.1
	Rs	67.2	68.9	72.5	34.7
	Sc	89.8	93.3	95.3	63.8
Edge-yolo	Cr	64.6	27.9	45.0	16.5
	In	77.7	77.8	**83.5**	44.7
	Pa	83.5	94.6	**94.4**	61.2
	Ps	75.2	82.1	89.9	56.4
	Rs	73.0	72.2	**78.7**	39.8
	Sc	75.4	88.5	91.4	57.0

For further comparison between YOLO11 and Edge-YOLO, we visualized both detection results and feature maps, as illustrated in Fig. 8 and Fig. 9. The visualization analysis

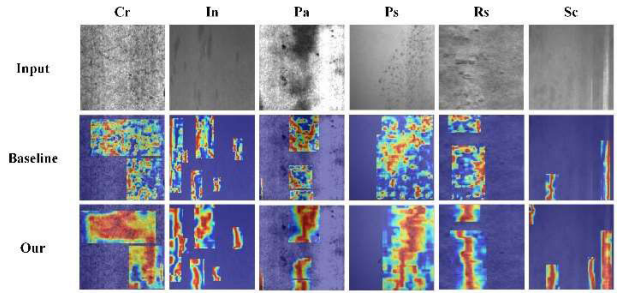


FIGURE 8. XGrad-CAM visualizations performed on the NEU-DET dataset.

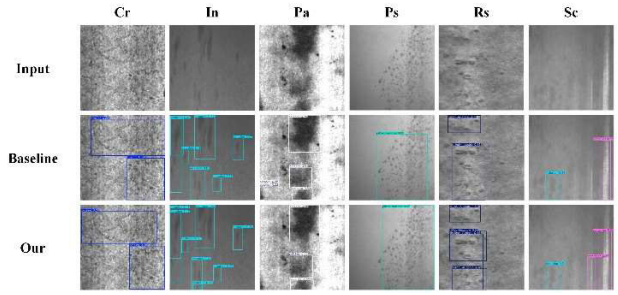


FIGURE 9. Some detection results of different models on the NEU-DET dataset.

provides qualitative insights into the performance differences between the two models.

To validate model effectiveness, we compared our approach with both general-purpose object detection models and specialized industrial defect detection methods. The comparison includes Classical two-stage detectors represented by Faster R-CNN, which achieves a moderate mAP of 70.8%, single-stage detection models including the YOLO series (YOLOv5, YOLOv7, YOLOv8, YOLOv9), with YOLOv9 reaching 78.9% mAP, and recent state-of-the-art defect detection models specifically designed for industrial scenarios, such as RT-DETR, FC-DETR, LDE-YOLO, and EA-YOLO. All experiments were conducted under identical hardware configurations and training protocols to ensure fair comparison. To eliminate the influence of transfer learning and evaluate the true learning capability of each model, all networks were trained from scratch without utilizing pre-trained weights. The metrics used for evaluation include mAP@0.5, Parameters M (million), and GFLOPs, providing a comprehensive assessment of both accuracy and computational efficiency. The results are summarized in Table 4, which demonstrates Edge-YOLO's competitive performance in both accuracy and computational efficiency against both general-purpose detectors and specialized defect detection approaches.

To evaluate the model's generalization capability, we performed detection experiments using YOLO11 and Edge-YOLO on both GC10-DET and GEAR-DET datasets, with results presented in Table 5. For further comparison between YOLO11 and Edge-YOLO, we visualized both detection results and feature maps, as illustrated in Fig. 10, Fig. 11, Fig. 12 and Fig. 13. The experimental results

TABLE 4. Defect detection results on NEU-DET dataset.

Methods	mAP@0.5/%	Parameters/M	GFLOPs
FasterR-CNN[34]	70.8	41.3	83.4
SSD[11]	71.1	24.5	30.6
RetinaNet[10]	63.2	36.4	74.5
YOLOX[14]	70.3	8.9	26.8
YOLOv5[23]	75.0	7.0	15.8
YOLOv7[18]	73.2	36.5	103.2
YOLOv7-tiny[18]	68.6	6.0	13.1
YOLOv8[22]	76.8	11.1	28.4
YOLOv9[19]	78.9	25.5	102.8
YOLO11[22]	78.4	2.46	6.3
RT-DETR[35]	73.5	28.5	100.6
FC-DETR[36]	83.7	56.5	143
LDE-YOLO[37]	80.8	22.2	65.6
EA-YOLO[38]	81.1	8.9	9.3
DsP-YOLO[7]	80.4	11.1	28.5
our	80.5	2.28	1.9

demonstrate significant performance improvements by Edge-YOLO on both datasets.

On the GC10-DET dataset, Edge-YOLO improved mAP@0.5 from 65.5% to 69.0%, representing a 3.5 percentage point increase. More notably, on the GEAR-DET dataset, Edge-YOLO substantially improved mAP@0.5 from 63.3% to 73.6%. This significant performance enhancement indicates that our proposed edge enhancement strategy exhibits particular advantages in gear component defect detection tasks, likely due to the distinctive edge characteristics typically present in gear surface defects.

These cross-dataset experimental results demonstrate Edge-YOLO's robust generalization capability, particularly across diverse industrial defect detection scenarios. The model not only adapts to different defect types and surface characteristics but also significantly reduces computational overhead while maintaining high detection accuracy. This combination of performance and efficiency holds substantial practical value for industrial applications.

2) Ablation Study

To explore the contribution of each component in Edge-YOLO, we conducted comprehensive ablation experiments on the NEU-DET dataset. The results are presented in Table 6, with six configurations:

- Baseline: Original YOLO model
- EdgeS: Baseline with Edge-Sensitive module only
- C3E2: Baseline with Cross-Stage-Partial Edge Enhancement module only

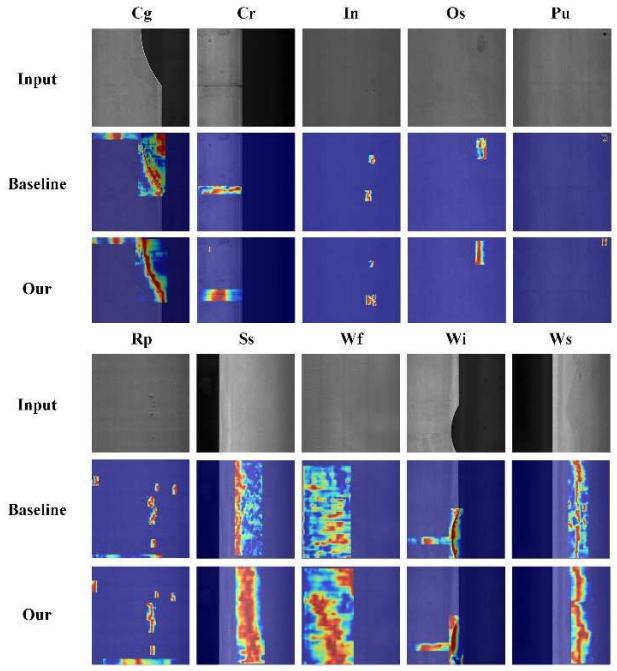


FIGURE 10. XGrad-CAM visualizations performed on the GC10-DET dataset.

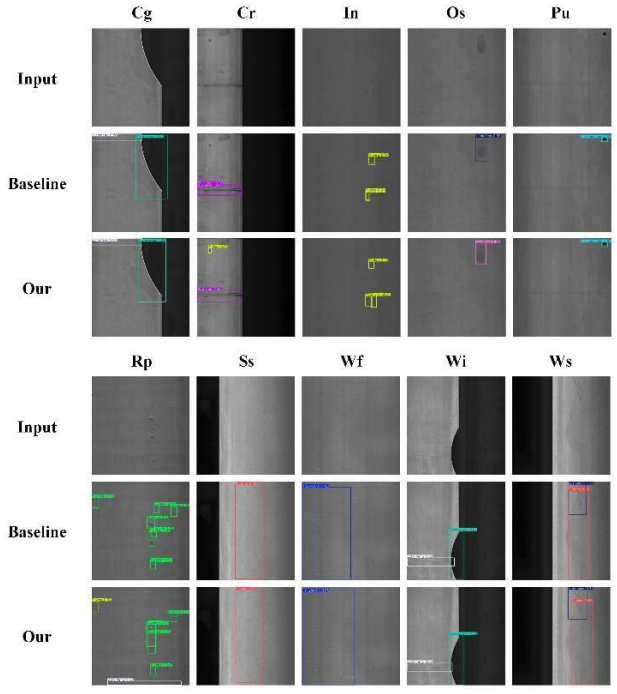


FIGURE 11. Some detection results of different models on the GC10-DET dataset.

- MSDC: Baseline with Multi-Scale Dilated Convolution module only
- Edge-backbone: Baseline model with edge-enhanced backbone only
- Edge-YOLO: Complete Edge-YOLO architecture with all improvements

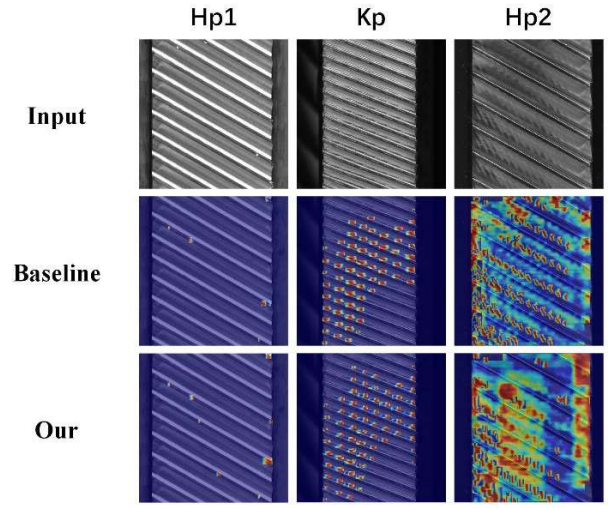


FIGURE 12. XGrad-CAM visualizations performed on the GEAR-DET dataset.

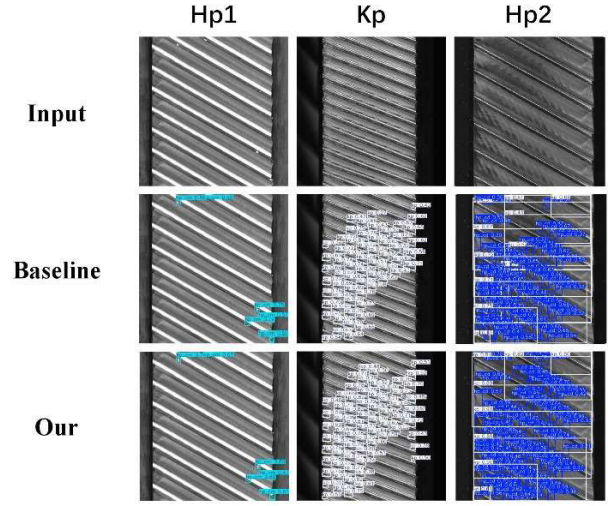


FIGURE 13. Some detection results of different models on the GEAR-DET dataset.

TABLE 5. Detection performance of YOLO11 and Edge-YOLO on GC10-DET and GEAR-DET datasets.

	Methods	mAP@0.5/%	Parameters	GFLOPs
GC10-DET	Baseline	65.5	2584102	6.3
	Edge-yolo	69.0	2387806	1.9
GEAR-DET	Baseline	63.3	2582737	6.3
	Edge-yolo	73.6	2386441	1.9

The individual module analysis shows that the EdgeS module alone improves mAP@0.5 to 79.1% while significantly reducing computational complexity to 2.1 GFLOPs. The C3E2 module achieves 78.1% mAP@0.5 with 6.4 GFLOPs, and the MSDC module reaches 77.3% mAP@0.5 with

6.3 GFLOPs. When these modules are integrated into the Edge-backbone, the performance improves to 80.1% mAP@0.5 while reducing GFLOPs to 2.2. The complete Edge-YOLO architecture further enhances performance to 80.5% mAP@0.5 while achieving the most efficient configuration with only 1.9 GFLOPs and 4.92MB model size. These results demonstrate not only the effectiveness of our edge enhancement strategy but also how each component contributes to the overall performance improvement, with the EdgeS module showing particularly strong individual performance in balancing accuracy and efficiency.

TABLE 6. The ablation experiments on NEU-DET.

Methods	mAP@0.5/%	Parameters	GFLOPs	Size (MB)
Baseline	78.4	2583322	6.3	10.5
EdgeS	79.1	2593386	2.1	5.28
C3E2	78.1	2530306	6.4	5.18
MSDC	77.3	2730778	6.3	5.55
Edge-backbone	80.1	2687826	2.2	5.47
Edge-yolo	80.5	2387026	1.9	4.92

VI. DISCUSSION AND CONCLUSION

A. DISCUSSION

The experimental results demonstrate Edge-YOLO's effectiveness across different industrial scenarios. The model's strong performance in detecting gear component defects particularly highlights its capability in handling complex industrial surfaces. The significant reduction in computational complexity (69.8%) while maintaining competitive accuracy suggests that our edge enhancement strategy effectively balances performance and efficiency.

While Edge-YOLO shows promising results, it's important to acknowledge several limitations:

- Performance under Extreme Conditions: The model's accuracy may decrease under severe lighting variations or highly reflective surfaces, which are common in industrial environments
- Computational Trade-offs: While we achieved significant reduction in computational complexity, this may impact the detection of extremely small defects in high-resolution images
- Real-time Processing Constraints: On certain edge devices, the processing speed might be affected when handling high-resolution images or multiple concurrent detection tasks
- Dataset Dependency: The model's performance heavily relies on the quality and diversity of training data, which might limit its generalization to novel defect types

These limitations suggest several promising directions for future research:

- Development of adaptive preprocessing techniques to handle extreme lighting conditions
- Investigation of more efficient feature fusion mechanisms for different defect types
- Enhancement of the architecture to better handle high-resolution images
- Integration of semi-supervised learning approaches to reduce dependency on large labeled datasets
- Optimization for specific edge computing platforms

B. CONCLUSION

This paper presents Edge-YOLO, a novel network architecture that enhances industrial defect detection through innovative edge-aware feature extraction. By redesigning the backbone network with edge-sensitive mechanisms, our approach effectively addresses the challenges of small-scale defect detection while maintaining computational efficiency. The edge-enhanced backbone, implemented through carefully designed modules (EdgeS, C3E2, and MSDC), demonstrates the effectiveness of our edge-centric design philosophy. Comprehensive evaluation across three industrial datasets demonstrates significant improvements in detection accuracy while reducing computational complexity by 69.8%. These results establish Edge-YOLO as a practical solution for industrial defect detection applications where both accuracy and computational efficiency are crucial.

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